

Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	6 September 2026
Team ID	Individual Project
Student Name	Morla Jahnavi
Project Name	An Employee Requests Installation of Licensed Software through the Service Catalog
Maximum Marks	4 Marks

Technical Architecture:

The proposed solution uses the ServiceNow cloud platform to manage the complete licensed software request lifecycle. The architecture connects the employee request interface, Service Catalog, Flow Designer automation, approval process, Catalog Task fulfillment, notifications, and ServiceNow data records. The architecture below identifies the main application processes, cloud boundary, user interfaces, organizational roles, and data storage components.

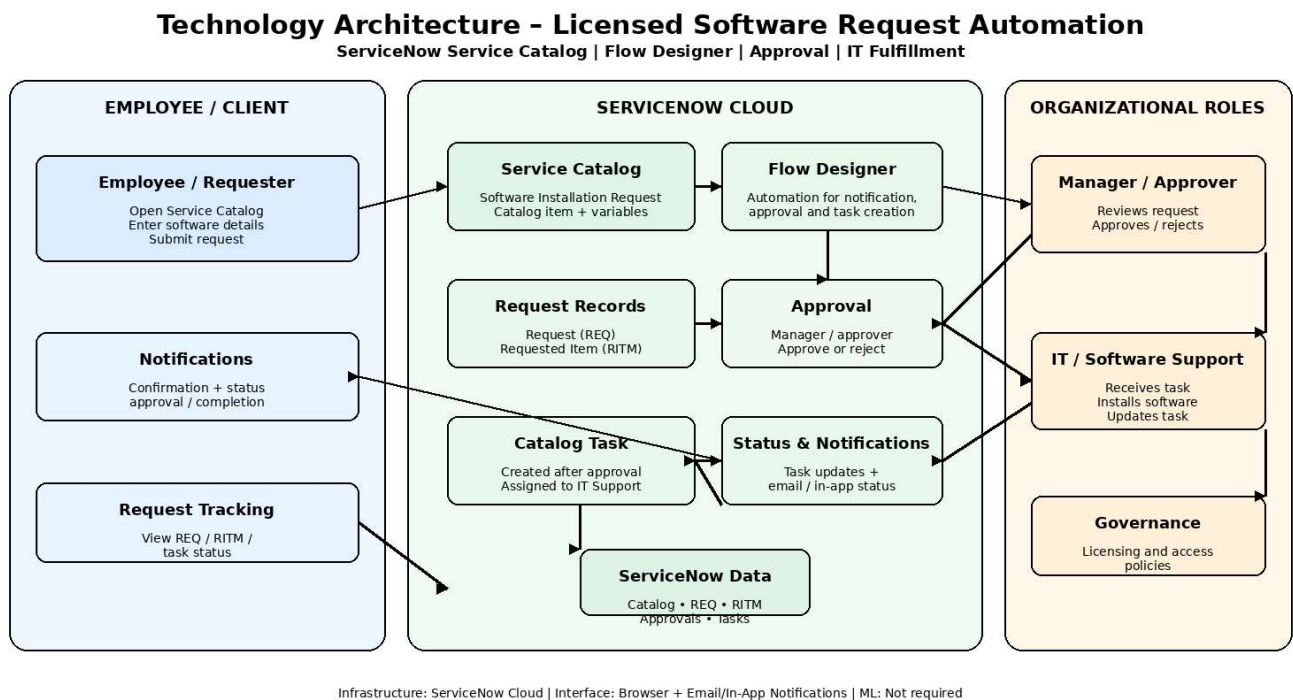


Figure 1: Technology Architecture for Licensed Software Request Automation

Architecture Description

The system uses ServiceNow to manage software installation requests.

1. Employee submits a software request through the Service Catalog.
2. ServiceNow creates a REQ/RITM and sends a notification.
3. The request goes to the Manager/Approver.
4. After approval, a Catalog Task is assigned to the IT Support Team.
5. IT installs the software and updates the task.
6. The employee receives the status and completion notification.

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1	User Interface	Employee submits and tracks the software request.	ServiceNow Service Portal / Service Catalog
2	Application Logic-1	Captures software details and creates the request.	ServiceNow Service Catalog
3	Application Logic-2	Automates notification, approval routing and task creation.	ServiceNow Flow Designer
4	Application Logic-3	Handles approval and fulfillment activities.	ServiceNow Approval Management + Catalog Tasks
5	Database	Stores request, approval and task information.	ServiceNow platform tables
6	Cloud Database	Managed cloud data storage for the ServiceNow instance.	ServiceNow Cloud Platform
7	File Storage	Stores supporting documents or request attachments when needed.	ServiceNow Attachments
8	External API-1	No external business API is required for the core workflow.	Not Applicable
9	External API-2	Notification interface for request status communication.	ServiceNow Email / Notifications
10	Machine Learning Model	No ML model is required for this workflow.	Not Applicable
11	Infrastructure (Server / Cloud)	Hosts application, workflow and data services.	ServiceNow Cloud Platform

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	No separate open-source application framework is required; the solution is configured within ServiceNow.	ServiceNow platform capabilities
2	Security	Role-based access and controlled permissions protect request, approval and fulfillment information.	ServiceNow Roles / ACLs / HTTPS
3	Scalable Architecture	Catalog, workflow, approval and fulfillment components can be reused for additional software request types.	ServiceNow Service Catalog + Flow Designer
4	Availability	The solution is hosted on the ServiceNow cloud platform and is accessed by authorized users through the platform.	ServiceNow Cloud
5	Performance	Automated workflow reduces manual coordination and moves requests through defined processing stages.	ServiceNow Service Catalog + Flow Designer

Architecture Summary:

The architecture keeps the software-request lifecycle within ServiceNow. Employees use the Service Catalog, Flow Designer automates the workflow, approvers make the required decision, and the IT Support team completes the Catalog Task. ServiceNow records provide centralized tracking. No machine-learning model or external business API is required for the core solution.